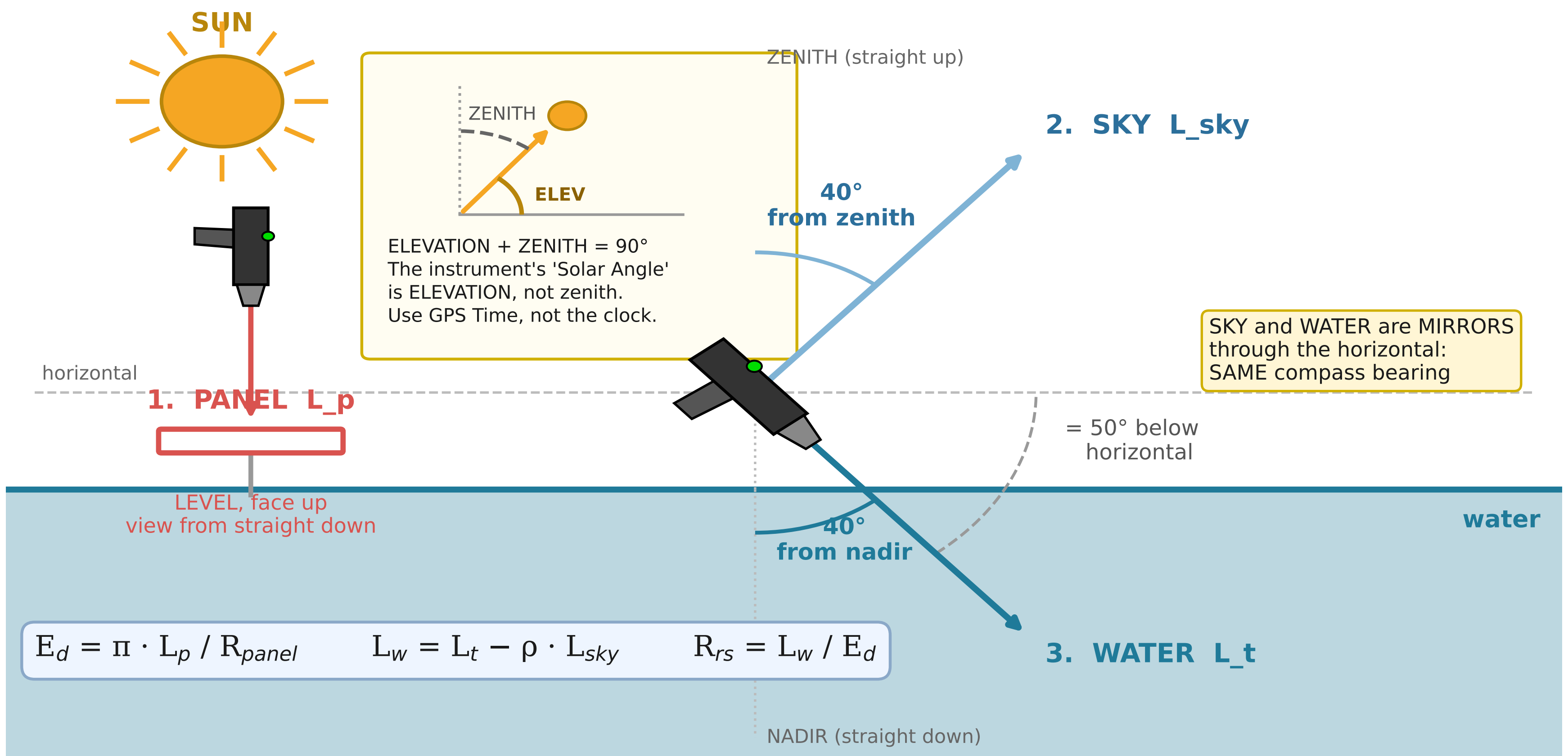
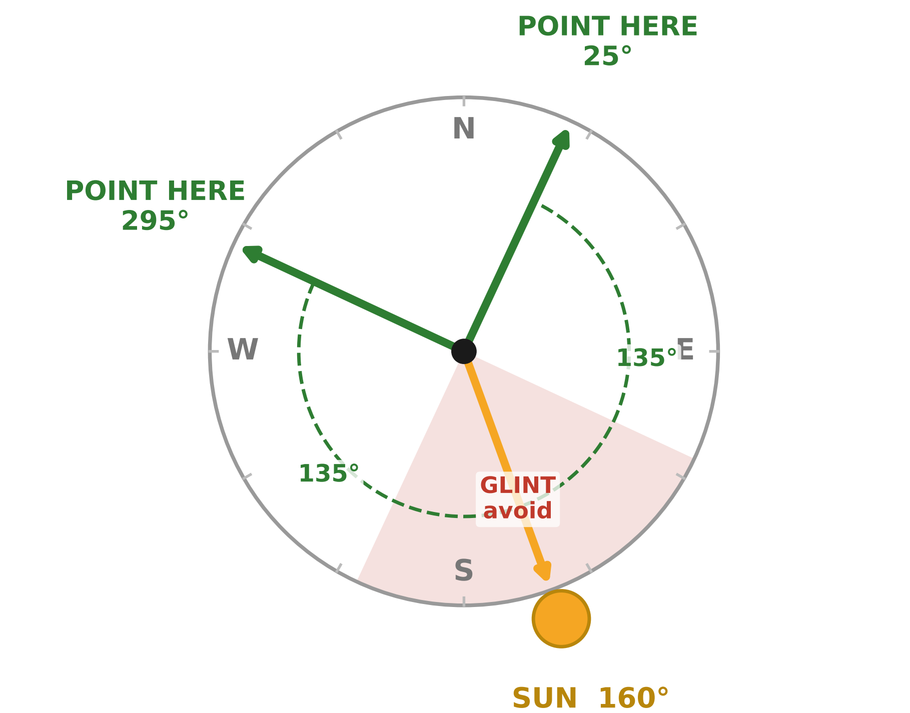
ABOVE-WATER R_{rs} — three-scan field method · Spectral Evolution NaturaSpec Plus

Geometry after Mobley (1999): view 40° from nadir, 135° in azimuth from the sun. Sky scan is the MIRROR of the water scan.

A. SIDE VIEW — the three scans and their angles



B. PLAN VIEW — which way to point (looking down)



Both bearings are 135° in azimuth from the sun. Pick whichever is clear of the boat, its shadow and its wake. The GUI computes these for you from latitude, longitude, date and time.

FIRST DECISION AT EVERY STATION: WHAT SKY IS IT?

CLEAR / MOSTLY CLEAR

Standard method. 40° / 135° from the sun.
rho = 0.028 only below ~5 m/s (see WIND).

UNIFORM OVERCAST → GO

NO SUN GLINT — better than clear in that one way.
rho stable, barely wind-dependent. 135° now meaningless:
keep 40°, pick bearing to clear the boat. NO BRDF. See page 2.

BROKEN / PATCHY CLOUD → STOP

THE WORST CASE, worse than either extreme. E_d changes
between panel and target scan; the error is unbounded and
invisible. Wait for it to settle, or use the E_d sensor (page 2).

1. PANEL

White reference panel (Spectralon),
LEVEL and face up, in full sun.
No shadow on it, yours included.
View from straight down.
Do not touch the white face.

DARWin: take as the REFERENCE.
Re-take every ~10 min, and the
moment the light changes.

2. SKY

Sky at 40° from ZENITH
= 50° ABOVE horizontal.

SAME compass bearing as the
water scan. Not the opposite
one. This is the single most
common field mistake.

DARWin: TARGET scan. Save.

3. WATER

Water at 40° from NADIR
= 50° BELOW horizontal,
135° in azimuth from the sun.
FROM A BOAT OR PIER use 90°:
135° looks back at the hull
or its shadow (IOCCG v3.0).
No glint, foam or wake.

DARWin: TARGET scan. Save.
Repeat 2 and 3 five times.

RECORD · WIND

WIND sets ρ, the largest error in
the whole measurement. NO
ANEMOMETER NEEDED — read it
off the water (Beaufort):

0–2	< 3 m/s	mirror, then wavelets, none breaking	ρ ok
3	3.5–5	crests break, SCATTERED whitecaps	limit
4+	> 5.5	FREQUENT whitecaps	INVALID

→ WHITECAPS APPEARING = the
edge of where ρ = 0.028 holds.

Also record: wind DIRECTION,
a photo, cloud / sun obscured,
panel reflectance (0.99?), and
clear or turbid water
turbid → NOT nir_zero

CHECK ON THE SPOT

⚠ THE CLOCK MAY BE WRONG.
Use GPS Time for solar
geometry. On a real file the
clock read 05:56 with the
sun below the horizon.

R_rs positive across 400-700?
negative = over-subtracted
Peak where the water looks?
Replicates on top of each other?
R_rs(443) order 1e-3 to 1e-2?

FOV × range = footprint. An 8°
lens at 3.8 m sees ~0.5 m of
water; the GUI plots it.

Copy the .sed files off tonight.

ABOVE-WATER R_{rs} · page 2 — CLOUD, THE E_d SENSOR, AND ABSOLUTE PRODUCTS

Overcast is a usable condition. Broken cloud is not. The irradiance channel is what makes the difference.

UNIFORM OVERCAST — what changes

BETTER than clear sky:

- NO SUN GLINT. The specular sun beam is the dominant error in above-water work. Under thick cloud there is no beam, so the hardest thing to avoid goes away.
- rho is stable and barely wind-dependent. Its wind sensitivity comes from wave facets sampling DIFFERENT PARTS OF A NON-UNIFORM SKY. Under a uniform sky there is little to sample between, so facet tilt stops mattering. rho = 0.028 is defensible even in wind that would rule it out on a clear day.

WORSE:

- E_d much lower → raise integration time, more replicates
- No satellite match-up: it cannot see through cloud either
- DO NOT apply BRDF. The Morel f/Q tables describe a clear-sky field with a direct beam; overcast is all diffuse and those tables do not describe it.

PROTOCOL: 135° azimuth becomes meaningless — no sun to point away from. KEEP 40° from nadir. Choose the bearing purely to clear the boat, its shadow and its wake. Record 'overcast': nothing in the file records the sky state for you.

BROKEN CLOUD — the worst case

Worse than either clear OR fully overcast.

The panel method assumes E_d is THE SAME at the moment of the panel scan and the moment of the target scan. Under moving cloud it is not, and the error is unbounded and invisible in the output — nothing in the spectrum tells you the light changed between scans.

OPTIONS, in order:

1. Wait for the sky to settle, either uniformly clear or uniformly overcast.
2. Use the cosine-collector E_d channel (panel opposite). Measuring E_d WITH the target removes the time lag.
3. If you must proceed: shorten the panel-to-target gap to seconds, bracket each target between two panel scans, take many replicates, and RECORD that the sky was patchy so the data can be down-weighted later.

A photograph of the sky at each station settles arguments later that no number can.

THE E_d SENSOR PATH — use it when light moves

Fit the COSINE DIFFUSER and take an irradiance scan. DARWin writes an 'Irr. (Target)' or 'Irr. (Ref.)' column, and the software uses MEASURED E_d instead of inferring it:

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rrs_from_sed(water, sky, source='irradiance')
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Three error sources drop out at once:

- the panel-to-target TIME LAG → the broken-cloud fix
- the PANEL REFLECTANCE entirely (no 0.99 assumption, no certificate, no multiplicative bias)
- PANEL LEVELNESS — the collector defines the horizontal plane itself

Conditions: the irradiance channel must be calibrated in $W\ m^{-2}\ nm^{-1}$ consistently with the radiance channel, and the COLLECTOR MUST BE LEVEL for the same reason the panel was. Its cosine error behaves better under diffuse light than under a low direct sun, so overcast is where it is strongest.

Verified: recovers a known R_rs to 9 decimals and is exactly invariant to panel reflectance.

YOU HAVE A SPECTRORADIOMETER — use the absolute scale

Absolute calibration supports products a reflectance-only instrument cannot give you.

PAR — Photosynthetically Available Radiation, from measured E_d. A PHOTON flux, not an energy flux, because photosynthesis counts photons:

$$PAR = \int_{400}^{700} E_d(\lambda) \cdot \lambda / 119.6 \; d\lambda$$

E_d in $W\ m^{-2}\ nm^{-1}$, λ in nm → $\mu mol\ m^{-2}\ s^{-1}$

Full midday sun ≈ 2000 $\mu mol\ m^{-2}\ s^{-1}$. Heavy overcast is one to two orders lower. Under cloud that number IS the record of how much light the water column actually received.

nLw — normalised water-leaving radiance, $nLw = R_{rs} \times F_0$. This is the quantity satellite ocean-colour products are distributed in, so it is what makes your field spectrum directly comparable with them.

Both need the IRRADIANCE channel, not just reflectance. Take an E_d scan at every station even when the sky is clear.